

Nikolas R. Sanderson

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EDUCATION

Northeastern University, Boston, MA

Expected May 2027

Candidate for Honors B.S. in Computer Engineering • GPA: 3.967

- **Research interests:** robotic navigation and autonomy, simultaneous localization and mapping, optimization

American College of Thessaloniki, Thessaloniki, Greece

Sep. 2022 – Dec. 2022

Study abroad program

WORK & RESEARCH EXPERIENCE

Visiting Research Assistant, University of Michigan

May 2026 – Present

Robotic Exploration Lab • Advisor: Prof. Alan Papalía

- Developed a camera–LiDAR perception pipeline for automated as-built vs. as-designed inspection of ship outfitting within a \$6.2M University of Michigan–led U.S.–Japan shipbuilding automation program; produced semantically annotated 3D maps by lifting SAM2 segmentations onto point clouds and applying 3D instance segmentation to isolate components (pipes, cables, equipment) in cluttered ship interiors.
- Conducted site visits to active Japanese shipyards, gathering firsthand insight into real-world outfitting workflows, clutter, and sensing conditions to inform perception system requirements and digital twin design.
- Represented the perception team at a program review and technical workshop in Japan (July 2026), contributing technical content on the camera–LiDAR pipeline to materials presented to government, industry, and academic partners.
- Lead author on an extension to [1], amending sparse variable projection to handle robust costs and achieving $2\times$ faster runtimes than a state-of-the-art robust solver; further characterized behavior under scaling and implemented a GPU version of the method. Led to full journal submission seen in [3].
- Developing a technique to handle One Way Travel Time (OWTT) Localization in Long-Duration underwater missions via estimating the clock state in the factor-graph optimization problem through incorporating its own factor that models clock drift.

Research Assistant, Northeastern University

May 2024 – Apr. 2026

Institute for Experimental Robotics • Advisors: Prof. Hanumant Singh, Prof. Michael Everett, Prof. David M. Rosen

- Developed a high-speed off-road autonomous robot platform, leading key design decisions across mechanical, electrical, and software systems, including integration of a state-of-the-art LiDAR SLAM system (DLIOM); demonstrated capabilities to the Army Research Lab in support of program reviews of graduate student mentors funding streams.
- Worked on a large team to develop a sensor payload for NEUROAM (a large-scale multi-robot data collect), integrating GPS, IMU, LiDAR, and cameras with ROS2 Humble-based time synchronization; handled end-to-end mechanical, electrical, and software integration, including Jetson configuration for deployment across Unitree and Boston Dynamics Spot platforms.
- Developed certifiably correct factors for back-end optimization algorithms in simultaneous localization and mapping (SLAM) and incorporated them into GTSAM allowing the design of certifiable solvers to be reduced from months to days; led to submission [2]. Presented work at the 2025 International Conference on Robotics and Automation Workshop *Robots in the Wild*.
- Co-authored [1] a sparsity-preserving variable projection method for large-scale robotic perception optimization that handles gauge symmetries via a one-time preprocessing step to build a matrix-free Schur complement operator, delivering $2\times$ – $35\times$ runtime speedups across standard robotic perception benchmarks compared to current state-of-the-art; released an open-source C++ implementation.
- Performed peer review for the 2026 IEEE International Conference on Robotics and Automation.

R&D Engineering Co-op, Robert Bosch LLC

Jul. 2024 – Dec. 2024

- Extended an automotive component test bench to evaluate hardware (e.g., compressors), integrating pressure and temperature instrumentation and developing data-collection workflows for characterization testing.
- Designed and executed automotive component validation experiments, selecting sensors (pressure, temperature, current, voltage), configuring acquisition and controls, and analyzing logged data to quantify performance across operating ranges.

- Developed a laser endoscope and an automated closed-loop robotic LED illumination system for photodynamic therapy, microsurgery, and microscopy of cancer.

PUBLICATIONS

- [1] Alan Papalia, **Nikolas Sanderson**, Haoyu Han, Heng Yang, Hanumant Singh, Michael Everett. “Sparse Variable Projection in Robotic Perception: Exploiting Separable Structure for Efficient Nonlinear Optimization.” *IEEE International Conference on Robotics and Automation (ICRA)*, Vienna, Austria, May 2026.
- [2] Zhixin Xu, **Nikolas R. Sanderson**, Hanna Jiamei Zhang, David M. Rosen. “Certifiable Factor Graph Optimization.” *Currently Under Review at IEEE Transactions on Robotics (T-RO)*. arXiv:2603.01267. (2026)
- [3] **Nikolas R. Sanderson**, Andrew Fishberg, Haoyu Han, Heng Yang, Jonathan P. How, Hanumant Singh, Michael Everett, Alan Papalia. “SPARSER: Sparse Variable Projection by Exploiting Separable Structure in Robotic Perception.” *Currently Under Review at the International Journal of Robotics Research (IJRR)*. (2026)

Workshop Papers

- [W1] Zhixin Xu, **Nikolas R. Sanderson**, David M. Rosen. “Simplifying Certifiable Estimation: A Factor Graph Optimization Approach.” ICRA Workshop on *Robots in the Wild*, Atlanta, GA, May 2025.
- [W2] Zhixin Xu, **Nikolas R. Sanderson**, Hanna Jiamei Zhang, David M. Rosen. “Certifiable Factor Graph Optimization.” ICRA Workshop on *Frontiers of Optimization in Robotics*, Vienna, Austria, May 2026.

PRESENTATIONS

- [1] *Multiwell Plate Photodynamic Therapy Experiments Using a Robotic LED Illumination System*. Biophysics Seminar, Northeastern Department of Physics, Boston, MA, March 2024.
- [2] *Automated Open-Source LED Robotic System Using the Nelder–Mead Numerical Optimization Method*. RISE Conference, Northeastern University, Boston, MA, April 2024.
- [3] *Building a Buoyancy-Controlled Float to Increase Accessibility to Marine Research*. RISE Conference, Northeastern University, Boston, MA, April 2024.
- [4] *Presenting Mariana: NUWave’s New ROV*. Marine Advanced Technology Education (MATE) ROV World Championship, Kingsport, TN, June 2024.
- [5] *Simplifying Certifiable Estimation: A Factor Graph Optimization Approach*. ICRA Workshop on *Robots in the Wild*, Atlanta, GA, May 2025.
- [6] *Presenting Calypso: A Production-Ready ROV*. Marine Advanced Technology Education (MATE) ROV World Championship, Alpena, MI, June 2025.
- [7] *A Platform for High-Speed Off-Road Autonomy*. IER Day research presentation, Northeastern University, Boston, MA, November 2025.
- [8] *Scalable and Certifiable Robotic Perception*. IER Day research presentation, Northeastern University, Boston, MA, November 2025.
- [9] *Presenting Imugi: A Final*. Marine Advanced Technology Education (MATE) ROV World Championship, St Johns, NL, Canada, June 2026

AWARDS & FELLOWSHIPS

Finalist, Sears B. Condit Award

Spring 2026

1 student selected from the Electrical and Computer Engineering department (~835 students) for university-level nomination.

Bruce Pelzer Merit Scholarship

Spring 2026

\$5,000 scholarship; 3 recipients selected from the Electrical and Computer Engineering department (~835 students).

PEAK Summit — Undergraduate Research Fellowship

Spring 2026

\$2,000 fellowship to research and develop simultaneous and localization and mapping algorithms.

Guts & Glory Award — Marine Advanced Technology Education (MATE)

Summer 2025

Awarded to the team that overcame hardship with determination and resolve.

MIT Sea Grant — STREAM Grant

Spring 2025

\$10,000 to develop an ROV for marine science and exploration.

Honors Propel Grant — Northeastern University Honors College

Fall 2024

\$2,000 toward SEASTAR, an education program for marine robotics.

Promoting Electric Propulsion — Naval Society of Engineers

Spring 2024

\$7,000 to build an unmanned surface vehicle (USV).

Eta Kappa Nu Honor Society

Spring 2024

Top 25% of the Electrical and Computer Engineering class.

Physics Internship Award — Northeastern Department of Physics

Spring 2024

\$3,000 fellowship to develop an optimal position planning algorithm.

Tau Beta Pi Engineering Honor Society

Fall 2023

Top 12.5% of the College of Engineering class.

Northeastern University Honors College

Fall 2023

PEAK Base Camp Award — Undergraduate Research Fellowship

Summer 2023

\$750 fellowship to pursue an independent research project.

PEAK Ascent Award — Undergraduate Research Fellowship

Spring 2023

\$1,500 fellowship to research and develop a buoyancy-controlled float.

CLUB PROJECTS

NOAA Ocean Exploration Challenge, NURobotics

Jul. 2024 – Aug. 2024

- Evaluated multiple object detection models (Faster R-CNN, YOLOv8, DINO) on the dataset; YOLOv8 was selected and submitted to the National Oceanic and Atmospheric Administration and Marine Advanced Technology Education organization.

Remotely Operated Vehicle, NURobotics

Apr. 2023 – Present

- Contributed to the design and build of an ROV, developed by a 30-person interdisciplinary student team competing at the MATE ROV World Championship.
- Served as Electrical Team lead from Sep. 2025 - Jun. 2026; now serving as overall Project Lead Sep. 2026 - present.

Buoyancy-Controlled Float, NURobotics

Sep. 2023 – Present

- Built a depth-profiling float that regulates buoyancy with a stepper-driven lead screw and syringe, logging depth and pressure over a wireless link to an operator station.
- Served as lead of this project from Sep. 2024 - Jun. 2025.

CubeSAT, Project Horizon

Sep. 2025 – Apr. 2025

- Led simulation of magnetorquers characteristics for Attitude and Determination Control System (ADCS) of a 1U CubeSAT which was launched on July 2nd 2026.

SKILLS

Programming: Python, C++, C, Julia (Clarabel, COSMO, Mosek), MATLAB

Robotics: ROS2, GTSAM, Docker, Git/GitHub, Linux, NVIDIA Jetson

Software: KiCad, SolidWorks, Blender, LTspice

Hardware: Oscilloscope, multimeter, soldering, 3D printing

VOLUNTEER & COMMUNITY SERVICE

Roxbury Robotics — Boston, MA

Apr. 2023 – Present

NURobotics Intro Course Instructor — Boston, MA

Jan. 2025 – Apr. 2025

Coastal Beach Clean-Up — Winthrop, MA

Jan. 2025

Tutor for Systemic Flow — Boston, MA

Jan. 2023 – Apr. 2023

Tutor for Cambridge School Volunteers — Cambridge, MA

Aug. 2020 – Jun. 2022